

30/10/25

Attention and Self-Attention

Seq 2 Seq

- Gating
 - Attention
- vanishing gradient

Attention vector $O_1 = [0.5x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4]$ weighted sum of the i/p

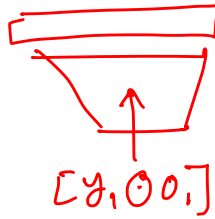
Attention distribution

0.5

0.2

0.2

0.1



unembedding layer

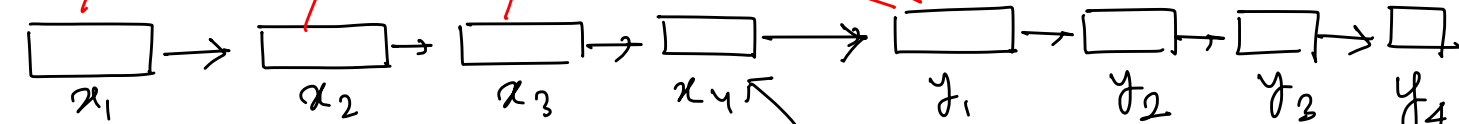
$x_1 \cdot y_1$

$x_2 \cdot y_1$

$x_3 \cdot y_1$

$x_4 \cdot y_1$

Attention score

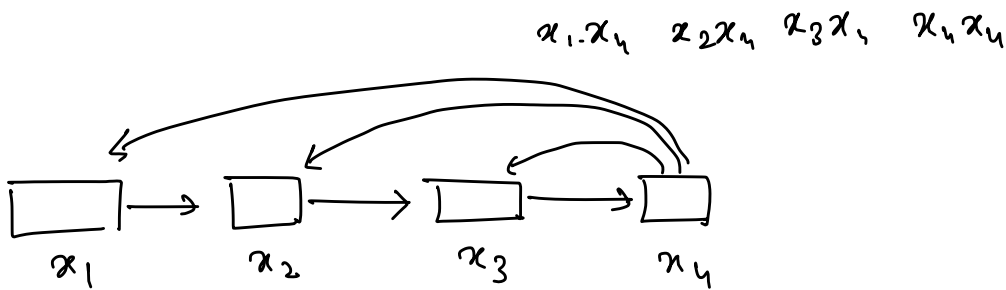


Encoder

Decoder

Conditional decoder

Bottleneck Problem



$$p_1 v_1 + p_2 v_2 + p_3 v_3 + p_4 v_4 = O_4$$

p_1

p_2

p_3

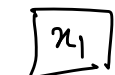
p_4

$x_1 \cdot K_1$

$x_1 \cdot K_2$

$x_1 \cdot K_3$

$x_1 \cdot K_4$



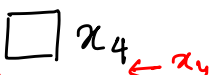
$\rightarrow K_1$
 $\rightarrow V_1$



$-K_2$
 $-V_2$



$-K_3$
 $-V_3$



$Q = x_4$
 $K = x_1, x_2, x_3$

	x_1	x_2	x_3	x_4
I	1	1	1	1
am	1	1	1	1
not	1	1	1	1
$well$	1	1	1	1

Attention Matrix

Query

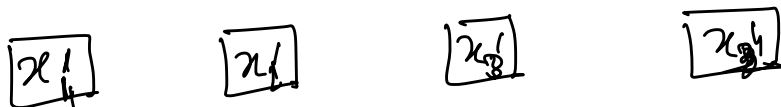
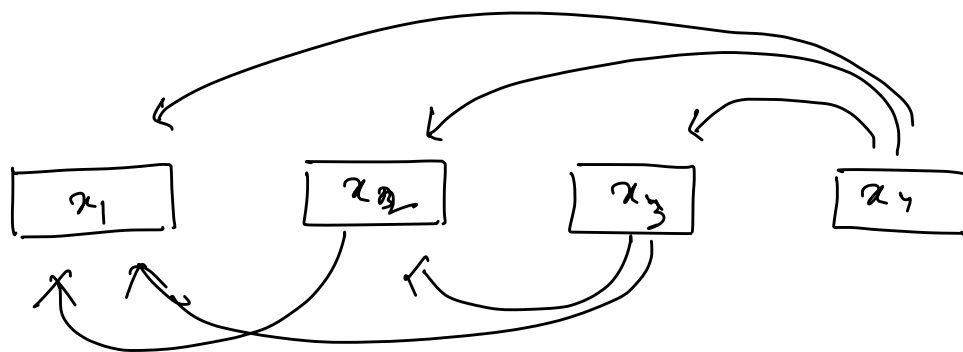
Key

Value

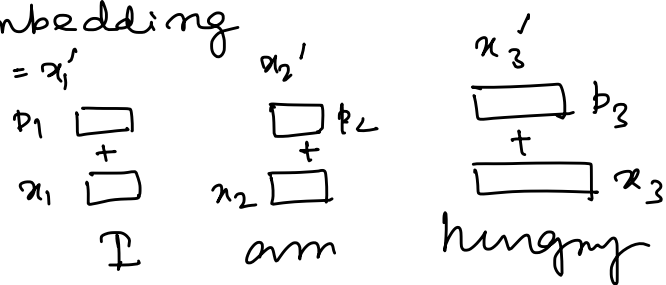
Attention vector is the weighted sum over all value vectors.

In case of normal attention, value and key vectors are the same.

Self-Attention



Positional embedding



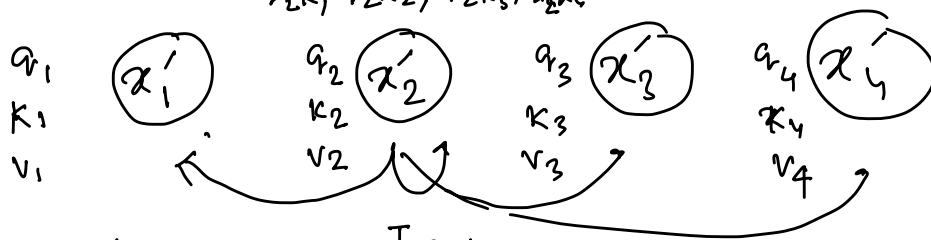
Query = W_q
Key = W_k
Value = W_v

$$O_1 = 0.7V_1 + 0.1V_2 + 0.1V_3 + 0.1V_4$$

$$\rightarrow \begin{matrix} 0.7 & 0.1 & 0.1 & 0.1 & 0.2 & 0.3 & 0.4 \end{matrix}$$

$$SM(q_1, k_1, q_1, k_2, q_1, k_3, q_1, k_4) \begin{matrix} 0.8 & 0 & 0 & 0.2 \end{matrix}$$

$$q_2, k_1, q_2, k_2, q_2, k_3, q_2, k_4$$



$$q_1 = W_q^T x_1'$$

$$k_1 = W_k^T x_1'$$

$$v_1 = W_v^T x_1'$$

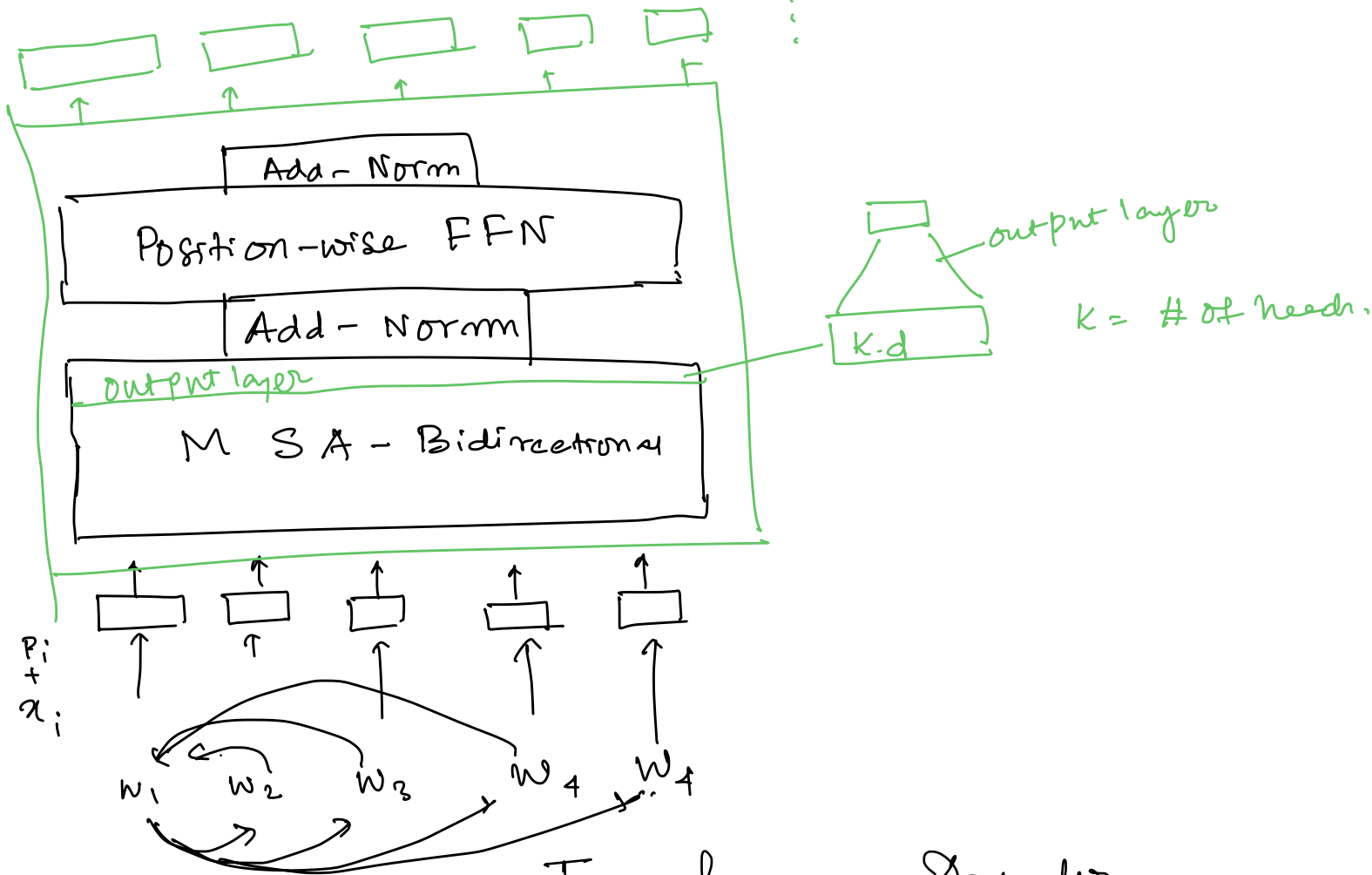
$$q_2 = W_q^T x_2'$$

$$k_2 = W_k^T x_2'$$

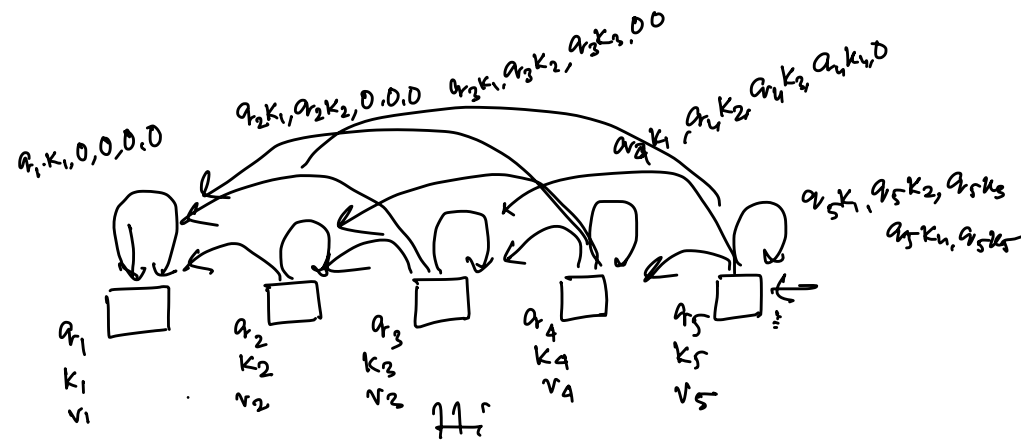
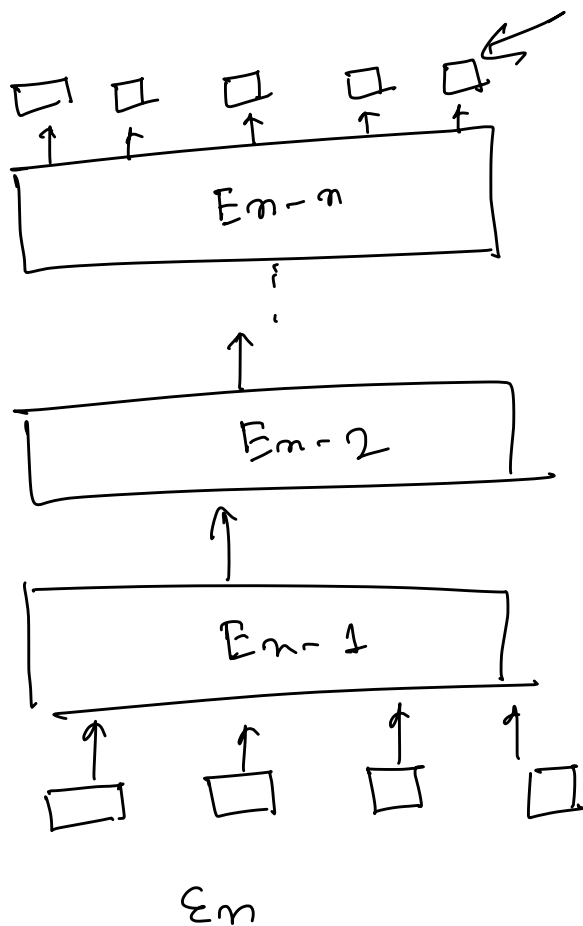
$$v_2 = W_v^T x_2'$$

03/11/25

Transformer Encoder



Transformer Decoder Masked MSA



q_1
q_2
q_3
q_4
q_5

k_1	k_2	k_3	k_4	k_5

q_1, k_1	q_1, k_2	q_1, k_3	q_1, k_4	q_1, k_5
q_2, k_1	q_2, k_2	q_2, k_3	q_2, k_4	q_2, k_5
q_3, k_1	q_3, k_2	q_3, k_3	q_3, k_4	q_3, k_5
q_4, k_1	q_4, k_2	q_4, k_3	q_4, k_4	q_4, k_5
q_5, k_1	q_5, k_2	q_5, k_3	q_5, k_4	q_5, k_5

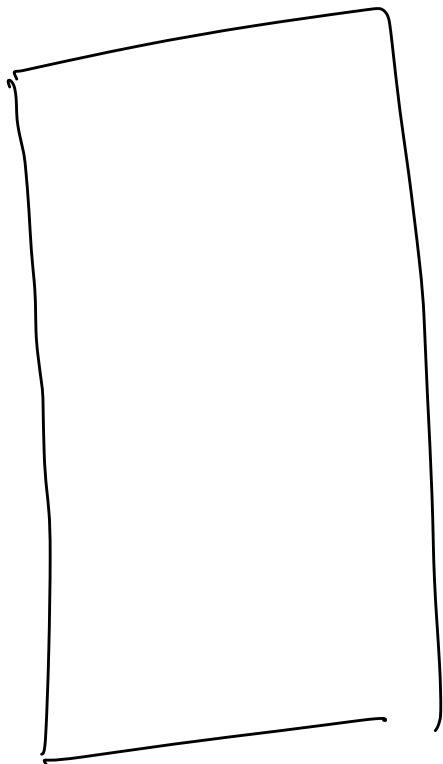
a_{1k_1}	a_{1k_2}	a_{1k_3}	a_{1k_4}	a_{1k_5}
a_{2k_1}	a_{2k_2}			
a_{3k_1}		a_{3k_3}		
a_{4k_1}			a_{4k_4}	
a_{5k_1}				a_{5k_5}

①

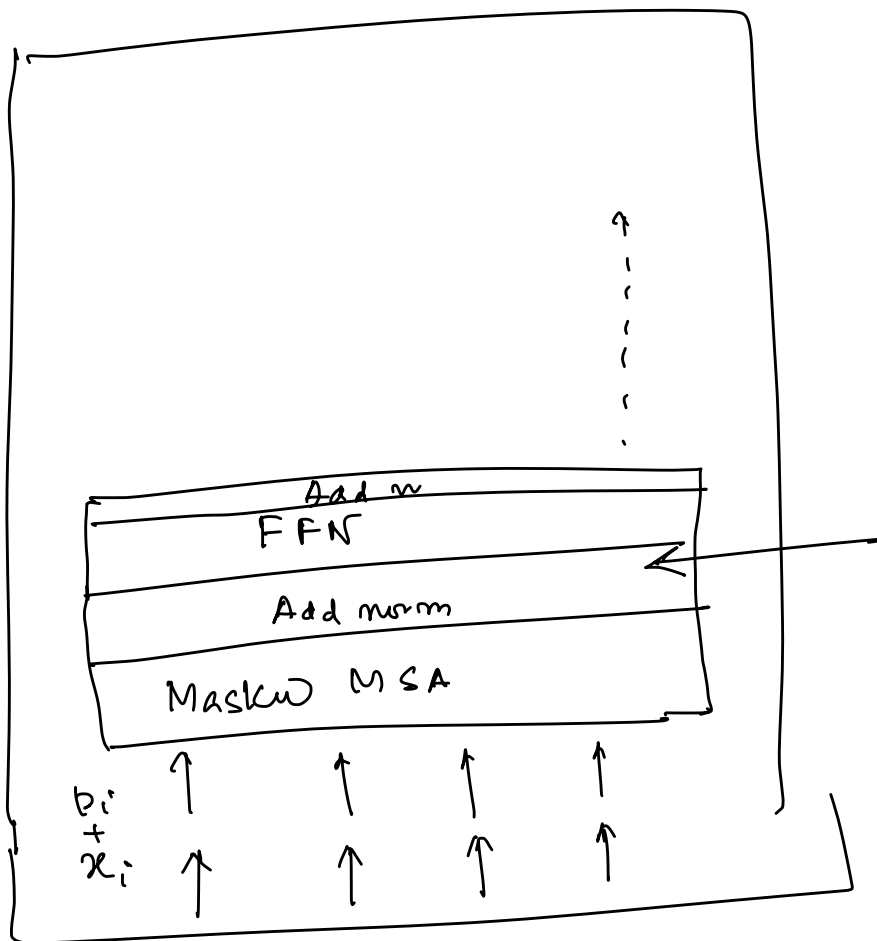
1	$-\infty$	$-\infty$	$-\infty$	$-\infty$
1	1	$-\infty$	$-\infty$	$-\infty$
1	1	1	$-\infty$	$-\infty$
1	1	1	1	$-\infty$
1	1	1	1	1

$$\frac{e^{a_{1k_1}}}{e^{a_{1k_1}} + e^{-\infty} + e^{-\infty} + e^{-\infty} + e^{-\infty}}$$

a_{1k_1}	$-\infty$	$-\infty$	$-\infty$	$-\infty$
a_{2k_1}	a_{2k_2}	$-\infty$	$-\infty$	$-\infty$
a_{3k_1}	a_{3k_2}	a_{3k_3}	$-\infty$	$-\infty$
a_{4k_1}	a_{4k_2}	a_{4k_3}	a_{4k_4}	$-\infty$
a_{5k_1}	a_{5k_2}	a_{5k_3}	a_{5k_4}	a_{5k_5}

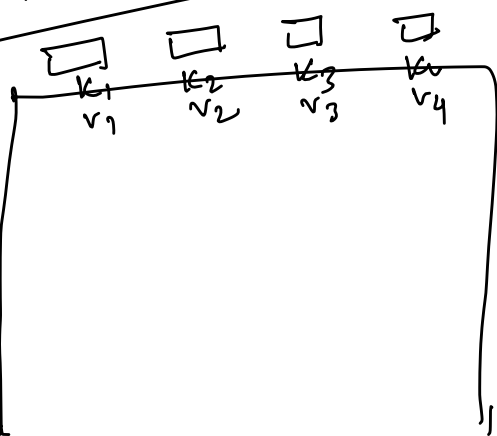


I_m



D_e

Cross-Attention



Q
 K
 V

BERT Pre-Training

Masked LM
objective
(MLM)
Self-Supervised
training

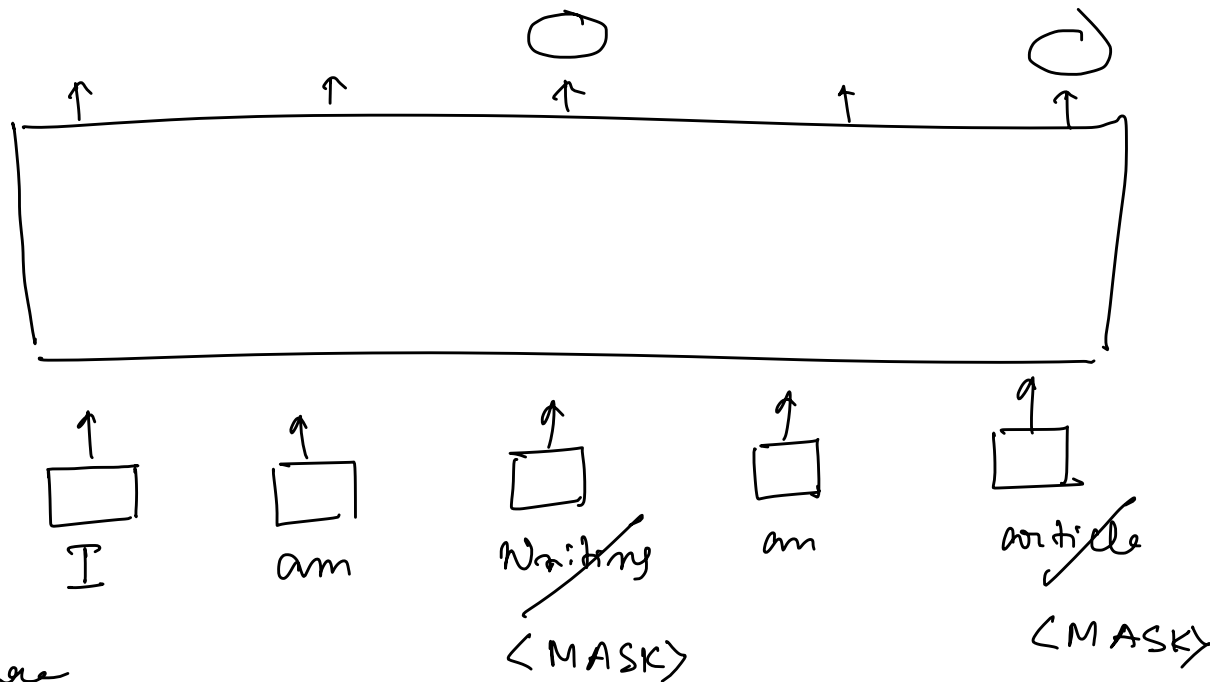
Intra
Sentence
dynamics

15%

- 50%
 <MASK>

- 80%
Another token

- 10% do not replace



Ob 2.: Next Sentence Prediction

